

# CS340 Final Project: Bias Mitigation

## Overview

In this project, you will open your imagination to a variety of methods to mitigate the bias from a benchmark model. The models used for evaluation extend from machine learning models to neural networks. In this project the dataset—ACSEmployment—is a rich real-world tabular dataset drawn from the American Community Survey. The task centers on predicting whether a person is employed based on a mixture of demographic, occupational, and educational attributes. However, beyond predictive accuracy, the critical challenge is to ensure that the model does not propagate or amplify societal biases—particularly those related to sensitive attributes such as sex (SEX) and race (RAC1P).

*To this end, you will be given a dataset and a sample code to mitigate the aforementioned bias using a hybrid of unrestricted methods. The anatomy of the task are generalized as follows, and*

- **Dataset and Benchmark:** Process the dataset, run the sample code and baseline model, and identify various types of bias. **(5 points)**
- **Experiment and report:** Modify on the pre-processing phase, the model structure, the training phase, etc., to obtain a bias-mitigated model and results. **(10 points)**
- **Performance:** Quantify the effectiveness of your mitigation. **(5 points)**
- **Presentation:** Compare the result of the new model with benchmark model and explain how they are different. **(10 points)**

As before, detailed steps will be given later, and you will then present your interesting and valuable insights by writing a report. **The assignment requires a submission of the *source code* and the *report*, and takes up a total of 30 points in your final grading.**

## Submission Guide

Please submit a zip file named `<StudentID-Name-FinalProject.zip>` to Blackboard, where the submitted zip includes two parts:

- **The project folder (.zip)**, which includes all the source code and other relevant files necessary for running the project.
- **A written report (.pdf)**, which is a complete and coherent description on how you evaluate bias and your insight on the results.

## How to get help

If you encounter any problems, please do not hesitate to reach out to the TA team by either sending posts on the Blackboard discussion board or making in-person appointments via QQ. We are here to help. Our ultimate goal is for all of you to acquire knowledge through proper training instead of overwhelming you :)

# Dataset

The ACSEmployment dataset is one of several datasets as an improved and more representative alternative to the widely used UCI Adult dataset. It is based on the 2018 American Community Survey (ACS) Public Use Microdata Sample (PUMS), a large-scale, nationally representative survey conducted by the U.S. Census Bureau. In this project, we only selected data from California (CA) and Texas (TX). The dataset contains 646,917 records.

Each data point describes an individual from the 2018 American Community Survey records in California and Texas. The features include demographic, educational, household, disability, migration, citizenship, and ancestry-related variables, such as age, education level, marital status, relationship to householder, disability status, citizenship status, mobility status, military service status, **SEX**, and race. The target variable is **EMPLOYED**, a binary label indicating whether the individual is employed.

The ACSEmployment dataset is particularly useful for studying fairness and equity in predictive modeling because it is based on a large-scale, standardized government survey and includes attributes associated with protected or socially sensitive groups. In this project, we use **SEX** as the sensitive attribute and study whether an employment prediction model behaves differently for Male and Female groups.

| Columns  | Description                                                                                          |
|----------|------------------------------------------------------------------------------------------------------|
| AGEP     | Age                                                                                                  |
| SCHL     | Educational attainment                                                                               |
| MAR      | Marital status                                                                                       |
| RELP     | Relationship to householder (husband or wife, housemate or roommate, nursing home, group home, etc.) |
| DIS      | Disability recode                                                                                    |
| ESP      | Employment status of parents                                                                         |
| CIT      | Citizenship status                                                                                   |
| MIG      | Mobility status over the past year                                                                   |
| MIL      | Military service status                                                                              |
| ANC      | Ancestry recode                                                                                      |
| NATIVITY | Nativity status                                                                                      |
| DEAR     | Hearing difficulty                                                                                   |
| DEYE     | Vision difficulty                                                                                    |
| DREM     | Cognitive difficulty                                                                                 |
| SEX      | Male or female                                                                                       |
| RAC1P    | Recorded detailed race code                                                                          |
| EMPLOYED | Binary employment label                                                                              |

Table 1: Description of dataset

# Now follow the steps

## 1. Process the dataset, run the benchmark model and analysis the result (5 points)

The provided dataset is in the `<acsemployment_2018_ca_tx.csv>` file. The baseline model code is provided in `<benchmark.ipynb>`, which contains the full training and prediction pipeline for the benchmark model. To ensure consistency across all submissions, do not modify any part of the data processing or the benchmark model at this stage, including adding or removing data or altering the model.

In this project, we focus exclusively on **SEX** as the sensitive attribute. After running the model and evaluating the sample code, you will obtain a variety of metrics. Use these metrics to identify potential sources of bias in the benchmark model.

### [TODO]

1. Run `<benchmark.ipynb>` to train the baseline model and obtain prediction results.
2. Complete the tasks in `<benchmark.ipynb>`.
3. Using the result from prediction, evaluate and analyze the bias existed in this model. Write up your observation. And the metrics of evaluation can be either the metrics we have learnt from Assignment 2 or any other metrics you preferred.

## 2. Mitigate bias from a variety of methods. (10 points)

Bias mitigation techniques reduce the disparities among the groups and intersectional subgroups measured by the metrics. Methods of mitigating model bias and their limitation is stated below:

- **Post-processing techniques:** aim to reduce fairness bias during model inference. Those approaches enforce model predictions to follow the same distribution observed during training. These techniques, however require access to protected attributes during inference, which is not always available due to data scarcity or privacy reasons.

- **Dataset pre-processing techniques:** It includes methods such as balancing the distribution of data labels, down sampling and sample re-weighting to alleviate modelling bias to a certain extent (as you work on Assignment 3). However, research shows that data pre-processing and balancing datasets often have limited effect, compared with training inherently unbiased models. Apart from data balancing, one can also delete protected attributes from the training set or mask them. However, this is not sufficient. since protected attributes are often correlated with other attributes in the data.

- **Train-time techniques:** This sort of methods aims to combat fairness bias during model training. This can be accomplished using constraints based on adversarial loss, feature importance, decision boundary, or statistical dependence. Adversarial loss requires defining additional heads or constraints for a specific protected attribute. It maximizes the primary objective of a specific task while minimizing the model's ability to predict specific protected attributes. Constraints based on feature importance, on the other hand, heavily rely on the feature contribution score, which is not always reliable.

### Some reference that may help to consolidate your project:

- **[1] Fairlearn API**

([https://fairlearn.org/v0.10/user\\_guide/mitigation/](https://fairlearn.org/v0.10/user_guide/mitigation/))

This package includes metrics for assessing model bias, and algorithms for mitigating unfairness issue. It

also provides an excellent and comprehensive tutorial for bias assessment and mitigation pipeline. You are encouraged to use these ready-made packages to assist your homework.

## - [2] Ethics and AI : how to prevent bias on ML ?

(<https://www.kaggle.com/code/nathanlauga/ethics-and-ai-how-to-prevent-bias-on-ml>)

This is an amazing tutorial on bias identification and mitigation. This will be a great example on how you outline your report, as this tutorial is quite similar to what you need to do in this project, i.e., bias analysis - bias mitigation - compare between models. Notice that our project does not require such a workload, but we need to learn its clear workflow.

For demonstration, we provide an example on how you can mitigate bias, where codes and detailed explanation are provided in file <benchmark.ipynb>. MinDiff is one of the TensorFlow Model Remediation techniques used to balance error rates (or, in this project, the FNR) between demographic groups (Male and Female) by penalizing distributional differences during training.

We trained a MinDiff model and evaluated it in comparison with the benchmark model. Following this process, you are required to design your own bias mitigation approach. You are encouraged to apply any methods you have learned from Assignment 3 and this project—combinations, integrations, or novel adaptations of techniques are all welcome. Finally, evaluate your model and compare it against the benchmark model.

### [TODO]

1. Read the reference materials to gain a comprehensive understanding of existing bias mitigation efforts. Read the example code to see how you can do it.
2. Design your method and model to combat bias. You can take any approach of a combination of them without limiting to the methods we introduced above. For example, you can take the data re-weighting method together with a loss function constraint method. However, it is **not allowed** to copy the sample code, but you can use the method in the example as a baseline and make adjustments. For example, modify the network structure and adjust the loss function. (**Note:** We do not admit that you only modified the hyperparameters in the example.)
3. Evaluate and analyze bias in the new model. It needs to be analyzed at the same dimension as benchmark model.
4. Write the report. It should be clearly stated your design idea, implementation details, rationale, evaluation result for your model.

### Compare the result of the mitigated model to the benchmark model. (5 points)

Now you have the benchmark model and your redesigned debiased model in hand. At the end of this project, you need to compare their performance in both accuracy and bias.

### [Research]

1. Compare the result from your analysis of two models.
2. Analyze the tradeoff between accuracy and fairness. Can they be achieved at the same time? Or does achieving better fairness come at the compensate of model accuracy?
3. Strive to improve the model's performance to get more points.

**Note:** We focus on the final score of your method as the final performance. We ultimately designed 13 core evaluation metrics. The more metrics on which your model outperforms the benchmark model, the higher your score will be. We categorize performance into five levels based on this comparison.:

| Level | Number of metrics outperformed | Points   |
|-------|--------------------------------|----------|
| Lv 1  | $\geq 2$                       | 1 point  |
| Lv 2  | $\geq 4$                       | 2 points |
| Lv 3  | $\geq 6$                       | 3 points |
| Lv 4  | $\geq 8$                       | 4 points |
| Lv 5  | $\geq 10$                      | 5 points |

#### 4. Presentation (10 points)

Make a slide showing your entire project implementation. It should contain details on the experimental plan, design, results, and the advantages and disadvantages of your method.